Pure-Weyl gravity programme: executive summary for physicists

Last substantive update: 21 July 2026 — certified two-phase counterflow causal BV parent and fixed-charge clock obstruction, exact five-dimensional compact-Cauchy adjoint kernel, action-derived observer repair no-go, and symbolic-frequency black-hole pairing programme.

> **Research context.** Asger Alstrup Palm directs the programme and is the accountable human contact. AI systems perform substantial derivation, verification, literature mapping, and drafting. Results are offered as scoped claims with reproducible artifacts. Palm's DTU affiliation identifies his academic role and does not imply institutional endorsement.

The sixty-second version

Pure Weyl gravity is a four-dimensional, locally conformal, fourth-order theory of the metric. Relative to Einstein gravity it has additional local gravitational solutions, and conventional pole or oscillator analyses attach indefinite signs to them. The programme starts from the position that a propagator sign is evidence, but does not by itself decide whether the theory contains a physical ghost particle.

Four logically distinct objects must not be conflated:

1. solutions of the local field equations;
2. structures remaining after gauge, charge, constraint, and boundary reduction;
3. directions that survive nonlinear continuation and interactions;
4. quantum states and asymptotic particles with a physical probability rule.

Substantial parts of the first two layers are now exact. Complete free causal gauge complexes have been constructed on several declared backgrounds. On a compact Einstein–Maxwell/Weyl–Maxwell laboratory, the Einstein image and additional fourth-order axial and polar summands are explicit. The axial additional block is nonradical under the action-derived Lee–Wald form, so it cannot there be dismissed as determinant multiplicity or local gauge. The sign distribution also does not coincide with the simple slogan "Einstein healthy, extra ghostly." These are classical current results, not quantum norms.

The nonlinear answer is sector-dependent. In the certified finite harmonic space, a finite exponential-polynomial second-order correction exists exactly on the zero set of five stabilizer moment maps—global quadratic charge-balance conditions. Bounded quasiperiodic continuation obeys further polynomial-growth and resonant-shell conditions. Some pure additional directions are therefore obstructed, while balanced combinations can possess formal secular extensions. Formal continuation is not stability.

A changed classical theory has now passed a new constructive gate. Two positive phase fields share one auxiliary diagonal $U(1)$ gauge connection; their gauge-neutral relative phase supplies a clock while the diagonal phase is contractible. On one selected fixed-Q_{rel} Berger background, the action is exactly equivalent to the previously certified positive Berger clock system plus a contractible gauge summand. This gives a complete real cyclic 70-component BV parent with support-local advanced and retarded chain homotopies. It is the first compensator-successor model to pass the complete classical causal-parent gate. Its full fixed-charge physical cohomology, reduced inertia, interactions, observables and quantization are still being audited. An exact homogeneous stationary classification has also shown that the passing fixed-action point is isolated rather than an open phase, and that the same action has no stationary path to the round cylinder. The homogeneous trace block is positive and oscillatory there; this does not replace the full nonhomogeneous physical reduction. That fixed-charge reduction has now been performed in the relative global sector: fixing Q_{rel} and quotienting its shift removes the entire relative-clock Darboux pair. Thus the branch on which D becomes null contains no physical relative clock. The alternative unrestricted branch retains the clock pair, but raw D is then a charged physical symmetry. Its full health audit remains open.

At one Euclidean loop order, strict fixed-field-content pure Weyl gravity has a nonzero local BV anomaly. A tau-adic compensator extension makes that local anomaly exact and restores the Euclidean quantum master equation at the tested order, but it enlarges the BV complex and counterterm space: it is a changed theory, not proof that strict pure Weyl gravity is anomaly-free. A separate reduced free mode space admits Hadamard two-point functions with signs $(+,-,-)$. No positive full-BV state, asymptotic particle space, scattering theory, or unitarity theorem presently exists.

The current verdict is therefore neither "ghost removed" nor "theory ruled out." Additional classical branches survive several gauge, radical, and horizon tests, while both parities at one Schwarzschild fixture are removed by finite symplectic norm at infinity. Nonlinear and quantum calculations impose further real obstructions; the general asymptotic and full-BV physical quotients remain open.

Current verdict

Established	Partial	Open
Complete causal free complexes on several declared backgrounds	Relational clocks, causal redshift, and localized detector preparations	Invariant branch-resolved interactions
Explicit Einstein-image and additional fourth-order classical summands	Bounded nonlinear continuation on selected finite mode spaces	Complete bounded finite-support cone

Established	Partial	Open
Nonzero Lee–Wald pairings for selected additional directions	Schwarzschild horizon admissibility plus two-parity finite-norm Einstein selection at one fixture	Asymptotic Bach phase space and scattering
Exact nonlinear balance conditions and a strict local Euclidean one-loop anomaly	A two-phase counterflow changed theory with a certified causal BV parent but an exact fixed-charge clock obstruction; compensator restoration and reduced Hadamard/Krein two-point functions on other carriers	Health of the unrestricted charged-time counterflow branch, a full-BV state, Lorentzian QME, particles, and unitarity

The public construction map shows the dependency structure; the searchable residual atlas tracks individual mode families without identifying them across backgrounds.

Starting choices and distinct laboratories

The initial cylinder calculation makes choices, not conclusions:

Starting choice	Purpose	It does not imply
Strict four-dimensional pure-Weyl action	Stress-test locally conformal fourth-order gravity	Correctness, stability, unitarity, or phenomenological viability
Conformal cylinder $\mathbb{R} \times S^3$	An exact closed causal laboratory for the complete free gauge complex	The topology or boundary conditions of the observed universe
Free linearized vacuum theory	Separate propagation and reduction from interaction failures	Nonlinear closure, matter compatibility, particles, or cosmology
Diffeomorphism and local Weyl gauge symmetry	Remove the defining local redundancies	That every residual conformal generator is proper gauge

Starting choice	Purpose	It does not imply
Selected zero-charge derived sector	Test the proposed residual quotient after deriving its charge condition	That cylinder time translation D is universally gauge
Full BV–BFV gauge complex	Track fields, equations, ghosts, identities, constraints, boundaries, and pairings together	That a raw pole or local mode is a physical state

Here **BV–BFV** means the gauge-complex framework containing fields, equations, ghosts, constraints, and boundary data. A **residual quotient** is the further quotient by gauge symmetries left after local gauge fixing.

Different questions require different backgrounds:

Laboratory	Principal test
Conformal cylinder $\mathbb{R} \times S^3$	Complete free causal complex and selected residual reduction
Berger clock backgrounds	Clocks, relational observables, gravity–matter interactions, and the two-phase counterflow causal-parent candidate
$\mathbb{R} \times S^1 \times S^2$	Einstein/additional waves, Taub constraints, and resonances
Nariai	Causal transfer on a different curved background
Schwarzschild and the static spherical family	Horizons, thermodynamics, and black-hole perturbations
Euclidean S^4 and generic Euclidean mode spaces	Anomaly, Hessian, and determinant calculations

These are distinct laboratories. A result on one background does not automatically transfer to another. Every comparison is indexed by

`(theory, background, generator, phase space, boundaries, lifecycle).`

Four flagship results

A. Causal gauge systems and residual reduction

On the conformal cylinder, the complete free metric BV–BFV complex has exact retarded and advanced Green homotopies, transport between support conditions, and matching Green, current, and residual pairings. Thus the residual calculation is connected to the covariant field theory rather than being an isolated Fourier quotient.

In the selected closed, zero-charge derived sector, all fifteen residual $SO(4,2)$ generators are imposed as constraints. The vacuum and selected residual one-particle cohomology are acyclic. The first surviving classes are

$$H^4 = \mathrm{span}\{[W_+^2], [W_-^2]\}, \qquad G = I_2.$$

They are deformation or vertex classes, not graviton particles. The sector qualification is essential: D is charged on the unrestricted compact phase space and becomes gauge only on the Taub-zero derived sector. A Taub obstruction is a second-order global integrability obstruction; its moment-map form is a quadratic charge-balance condition.

The finite residual mode window is not an arbitrary-support replacement for the all-energy causal theory. An exact weight-five test finds 64 Einstein cohomology directions in the source and none in the selected target, so a D-equivariant unary deformation retract to that finite receiver cannot exist. This leaves the finite residual theorem intact, but blocks the proposed arbitrary-support transfer of the nonlinear operation and anomaly classes until an all-energy completed time-slice carrier is constructed.

The causal-transfer theorem also has Berger, flat-Minkowski, and curved Nariai consumers. On Nariai the complete repaired system retracts to the metric Bach endpoint with exact causal homotopies and one certified formal transverse background variation.

The newest Berger result concerns a genuinely changed action rather than a passive compensator. Two clock phases counterflow so that the diagonal compact gauge charge vanishes while the relative phase evolves. On the selected squashed background and fixed relative-charge leaf, the resulting theory has a complete 70-component action-derived causal BV parent. The exact fixed-action homogeneous stationary locus contains this positive solution as an isolated point and supplies no stationary route to the round cylinder. Raw D remains charged on the unrestricted parent. On the derived fixed-Q_{rel} fibre, quotienting the relative shift makes D null but also contracts the complete relative-clock Darboux pair: the physical clock dimension is zero. The next Classical gate is therefore the distinct unrestricted branch, where the clock survives but time translation is a charged physical symmetry. Positivity of the homogeneous trace block alone is not a full health theorem for that branch.

> **Immediate limitation.** This is not a universal removal of radiative gravitons, an open viable phase for generic backgrounds, or an asymptotic particle theorem.

B. Einstein-image and additional classical branches

On $\mathbb{R} \times S^1 \times S^2$, the complete linear Einstein–Maxwell tangent maps into the Weyl–Maxwell system. The resulting linear triangle and its relative space are explicit, including global and large-gauge endpoints. The triangle is not cyclic under the standard pairings: Einstein inclusion is not symplectic equivalence.

For generic axial $\ell \geq 2$, quotienting the Weyl–Maxwell solution space by the Einstein–Maxwell image leaves two exact additional polarizations. A direct four-dimensional Lee–Wald calculation makes this additional block nonradical with signature $(2, 0)$; the complete generic axial space has signature $(3, 1)$, with the negative direction on an Einstein-image master branch. "Einstein" and "positive," or "additional" and "negative," are therefore not synonymous on this compact classical space.

The polar operator independently contains the complete Einstein primary summand plus two additional summands on every declared physical fibre. Its action normalization is derived, but the direct polar Lee–Wald form remains open.

> **Immediate limitation.** Final residual descent, the ungauged polar Noether lift, causal boundary admissibility, and positive-frequency quantization have not been completed. A nonzero classical pairing is not a particle norm.

C. Nonlinear continuation, interactions, and resonance

Paper 13 classifies the second-order tangent problem on the certified finite harmonic Weyl–Maxwell space. A **finite exponential-polynomial correction** is a finite sum of $t^p e^{-i\Omega t}$ terms. Such a correction exists exactly when five stabilizer moment maps vanish. This is an exhaustive model-specific realization of the classical linearization-stability and moment-map framework, not a claim to have invented that general theory.

Bounded quasiperiodic continuation is stricter. It additionally requires the vanishing of positive polynomial-growth coefficients and adjoint pairings on resonant characteristic shells. On the complete global plus $(\ell = 2, k = 0)$ additional-primary space, the bounded cone reduces to the standard global generalized-zero cone: no nonzero additional-primary amplitude survives. Other balanced finite combinations can possess formal exponential-polynomial corrections with secular growth. The full bounded finite-support zero locus remains open.

The standard-branch calculation now extends uniformly to every $\ell \geq 2$ at a separately tuned compact momentum. Its exact resonance zero set consists of two mixed axial–polar sheets and two one-sided planes; after the global energy and momentum balances, the bounded cone is the origin plus the two mixed sheets on an explicit occupation interval. In a separate two-absolute-momentum enlargement, all 108 axisymmetric $L = 4$ and all 56 nonaxisymmetric $L = 1, 3$ branch-basis coefficients are exact and individually obstructed. This complete

164-coefficient basis census is not yet the arbitrary-amplitude zero variety: cross-column cancellations remain open.

The Berger interaction programme has exact cyclic q_2, q_3 tensors and a retained mixed gravity–Maxwell ℓ_3 representative on its declared mode space. That representative is an architectural result, not yet an invariant physical interaction. Once the fuller Maxwell ghost structure is included, the mixed interaction is cyclically removable through first jet. The second-jet image calculation is still active; no nontrivial cyclic deformation class or Einstein/additional/Maxwell mixing table is certified. The nonlinear symmetry that does close is $K_{\text{Berger}} = D - \omega R$, not raw D.

The Einstein–Maxwell-to-Weyl–Maxwell inclusion has also reached its first nonlinear obstruction. The common action-derived q_1, q_2, q_3 tensors pass their internal identities, but the strict arity-two inclusion defect is nonzero. Taub pairing rules out a support-local correcting f_2 on the full smooth periodic fixed-bundle carrier. Its global cohomological image is nevertheless exactly the five stabilizer charges. Locality therefore forces the relative receiver to be a Noether-current complex, with charges obtained only after integrating over a Cauchy slice. The canonical Hessian Green current and its local divergence identity are now exact on all fourteen physical rows. Stabilizer precomposition, cyclic-dual completion, Lee–Wald comparison, and recovery of all five integrated charges are the next gates.

> **Immediate limitation.** Exponential-polynomial continuation is not bounded stability, retarded solvability, Sobolev existence, an invariant interaction amplitude, or all-order nonlinear integration.

D. Quantum anomaly and black-hole endpoint tests

For strict fixed-field-content pure Weyl gravity, the classified local one-loop Euclidean BV anomaly has nonzero coefficients. The local Euclidean quantum master equation is therefore obstructed. In a tau-adic compensator extension, the anomaly becomes exact and the local equation is restored at that order. This second result changes the BV complex and counterterm space; it does not make strict pure Weyl gravity anomaly-free.

Separately, a reduced free cylinder mode space with Einstein-like, additional, and longitudinal families admits compatible Hadamard two-point functions—the required short-distance singularity structure—with indefinite signs $(+, -, -)$. These Krein signs are signs in a candidate state pairing, not a full BRST-positive Hilbert space. The free curvature-observable CCR algebra is also exact, but it has no certified compatible positive state. On $S^2 \times S^2$, a separate Euclidean spectral calculation now gives the exact longitudinal ghost Schur eigenvalue and a coupled exceptional-mode correction. It advances the determinant assembly but remains `EUCLIDEAN-SPECTRAL`, not a Lorentzian state or unitarity result.

The black-hole programme classifies the static spherical Bach-flat family and obtains a consistently normalized energy, Wald entropy, and first law on its declared domain. On Schwarzschild, the axial Einstein/Regge–Wheeler family and an additional Ricci-curvature family are explicit. The additional family has future-horizon-regular ingoing solutions. Controlled real-frequency fixtures give nonzero Einstein–additional and additional–additional horizon pairing, while the pure Einstein block is null in the pure-Weyl current.

Horizon analyticity alone does not force the additional family to vanish. But at $\ell = 2$, $\omega = 3/5$, the completed axial and polar metric reconstructions have divergent Einstein–extra and extra–extra slice norms, while the Einstein sector is finite (with an additional zero polar self-density in one sector). Thus finite symplectic norm at infinity selects exactly the Einstein sector in both parities at this fixture. This is a global phase-space selection, not a local horizon condition or a general all-frequency theorem.

> **Immediate limitation.** A symbolic-frequency theorem, general ℓ, summability, a complete asymptotically flat phase space, the exterior initial-boundary problem, complex frequencies, ringdown, and stability remain open.

Where the strongest criticism currently lands

The strongest criticism is correct on six points:

- D is not universally gauge; it is charged on the unrestricted compact phase space and is expected to be physical in ordinary asymptotic settings.
- Acyclic selected cylinder one-particle cohomology is not asymptotic unitarity or absence of radiative degrees of freedom.
- Einstein inclusion is not nonlinear closure, symplectic equivalence, or an exclusive sector of Weyl gravity.
- Additional classical branches survive selected gauge-radical and horizon tests, while finite symplectic norm excludes the extra branch in both parities at one certified Schwarzschild fixture.
- The Berger nonlinear generator is $K_{\mathrm{Berger}} = D - \omega R$, not raw D, and the retained interaction has no certified invariant branch meaning.
- The compensator repairs a changed local Euclidean theory; it does not remove the strict theory's anomaly.

What remains valid is correspondingly precise: the zero-charge condition is derived rather than assumed; the selected residual quotient is connected to the covariant causal complex; a healthy clock can coexist with total zero-charge D at linear fixed coupling; the compact additional axial space is nonradical; exact nonlinear balance and resonance obstructions exist; and the strict/extended quantum alternatives are computationally separated. The two-phase counterflow construction now supplies one changed-theory causal parent worth consuming before another classical architecture is introduced.

Authoritative result ledger

The rows below are the authoritative short formulations. Raw implementation milestones are excluded.

Scientific question	Current result	Lifecycle	Principal limitation / route
Free causal complex and residual cohomology	Complete cylinder causal complex; selected zero-charge residual vacuum and one-particle cohomology acyclic; $H^4 \cong \mathbb{C}^2$ contains deformation classes	**Certified on their distinct carriers**	The finite residual receiver is not an arbitrary-support all-energy retract; the latter has a rank-64 unary obstruction and requires completion; Papers 7–8
Is D gauge?	Charged on the unrestricted compact phase space; gauge on the Taub-zero derived sector	**Certified, sector-dependent**	Must be recomputed with boundaries and matter; status ledger
Clocks and redshift	Healthy original Berger clock and retarded Maxwell redshift fixture; the old apparatus repair is obstructed; the two-phase theory has a causal parent, but its fixed-charge quotient removes the relative clock	**Original observable certified / changed fixed-charge clock obstructed**	The unrestricted charged-D branch and a new action-derived observable remain open; Paper 9
Einstein image and additional branches	Linear inclusion, axial additional nonradical block, and polar primary decomposition certified	**Certified classically**	Polar current, residual descent, causal boundaries, and quantization open; Paper 92

Scientific question	Current result	Lifecycle	Principal limitation / route
Nonlinear tangent cone and resonances	Formal finite exponential-polynomial cone equals five-charge zero set; tuned standard bounded cone is classified for every $\ell \geq 2$; all 164 two-momentum basis coefficients are closed	**Formal cone and all-ℓ standard subcone certified; larger cone partial**	Arbitrary-amplitude two-momentum zero variety, causal, infinite-mode, and all-order problems open; Paper 13
Interaction and branch mixing	Retained mixed tensor is exact/cyclic; fuller ghost structure removes it through first jet	**Representative certified; invariant class open**	Second-jet image and branch-resolved operation not certified; Paper 11
Free quantum observables and reduced states	Curvature CCR algebra exact; separate reduced E/A/L Hadamard carrier has signs $(+,-,-)$	**Certified on distinct free carriers**	No full-BV positive state, particles, or interacting products; state report
One-loop anomaly	Strict local Euclidean QME obstructed; compensator extension restores changed theory at one loop	**Strict obstructed / extension restored**	No Lorentzian QME or unitarity theorem; Paper 12

Scientific question	Current result	Lifecycle	Principal limitation / route
Black holes	Static first law; additional axial and polar families reach the horizon, but finite symplectic norm at infinity selects the Einstein sector at the certified $\ell = 2, \omega = 3/5$ fixture	**Static certified / two-parity fixture classified**	Symbolic frequency, general ℓ, full scattering phase space, stability, and ringdown open; Paper 14
Asymptotic particles and unitarity	No theorem	**Open**	Requires asymptotic Bach phase space, positive physical state, scattering, and optical/unitarity tests

Highlights by audience

Higher-derivative gravity and unitarity

The compact current calculation separates extra roots from gauge and radical directions, while placing one negative classical direction on an Einstein image rather than the additional axial block. A potentially distinctive element is the common comparison of local modes, charge sectors, nonlinear obstructions, and reduced quantum signs. The decisive gate is a positive full-BV or asymptotic physical state, not another pole count.

BV/BRST and perturbative quantum field theory

The programme supplies complete classical gauge-complex inputs, a free causal curvature algebra, a classified one-loop local anomaly, and a compensator extension in which that anomaly is exact. A potentially distinctive element is the fail-closed separation of local algebraic, Euclidean spectral, reduced-mode, and Lorentzian causal claims. The open gate is a renormalized Lorentzian QME and a positive state on the full BV complex.

Lorentzian PDE and Green-hyperbolic complexes

Exact retarded/advanced Green homotopies transfer through cyclic differential contractions to fourth-order gauge endpoints on several backgrounds. A potentially distinctive element is constructive pairing transport through tractor/HPL reductions when no isolated central operator provides the whole causal object. The open gate is stability on a finite family of less symmetric globally hyperbolic backgrounds.

Conformal geometry and detour complexes

The tractor parent, BGG/detour endpoint, differential contraction, and Lorentzian Green homotopy are explicit in one large conformal-gravity example. A potentially distinctive element is the causal interpretation of the Bach operator as a compression of a larger gauge complex. The next gate is one second detour or higher-spin consumer and a background-family theorem.

Mathematical relativity and black holes

The compact programme gives explicit charge-fibre-dependent Taub obstructions and all-ℓ resonant refinements; the black-hole programme gives a static first law, Ricci-carrier factorization, horizon reach, and finite-norm Einstein selection in both parities at one fixture. Potentially distinctive elements are the exhaustive finite-mode cone and the pure-Weyl horizon/asymptotic pairing comparison. The gates are the joined two-momentum zero variety and a complete Schwarzschild exterior phase space.

Relational observables and clocks

A healthy Berger matter clock coexists with zero total D charge in one linear fixed-coupling sector, and a retarded Maxwell signal defines a gauge-invariant clock-slice redshift observable. A potentially distinctive element is nontrivial relational evolution with total presymplectic degeneracy. The next gate is one localized, backreacting emitter–receiver experiment with recoil and an apparatus bracket.

What would materially change the verdict

A major positive shift would be produced by any one of: a nondegenerate positive asymptotic Einstein sector with the additional family excluded by a physical boundary rule; a positive full-BV Hadamard state; a certified nontrivial branch-resolved interaction compatible with the constraints; or a localized backreacting relational measurement.

A major negative shift would be produced by: an unavoidable negative physical direction after residual and boundary reduction; a nontrivial Lorentzian quantum Cartan anomaly; failure of every admissible nonlinear Einstein-sector boundary condition; or an asymptotic additional branch with negative physical flux that cannot be consistently excluded.

Either outcome is valuable. The programme is designed to locate the first precise obstruction, not to protect a preferred interpretation.

What collaborators can use now

The repository can support a collaborator who wants to:

- audit one flagship theorem against an independent convention;
- supply a preferred operator, gauge, boundary condition, or background;
- compare a known result with a complete certificate chain;
- inspect one unresolved defect or deliberately broken verifier input;
- help construct the asymptotic Bach phase space;
- review the anomaly/BV argument or the nonlinear tangent-cone theorem;
- run an adjacent detour, higher-spin, black-hole, or observer problem through the existing symbolic infrastructure.

The programme is not asking a new collaborator to endorse the entire interpretation. It is asking them to attack one scoped claim with full access to its derivation and verifier.

Detailed active projects

Project	Current boundary	Exact next gate
Cyclic causal Green transfer	Abstract conditional theorem with Berger, flat, and Nariai consumers; one Nariai formal tangent	Extend to a finite background family or second detour theory.
Charge-fibre obstruction	Formal cone, all-ℓ tuned standard cone, and all 164 two-momentum basis coefficients certified	Classify the common amplitude zero variety.
Additional axial/polar current	Axial direct current and polar equation module certified	Compute the polar current, ungauged lift, residual descent, and physical boundary disposition.

Project	Current boundary	Exact next gate
Relational clock and light	Original global retarded redshift certified; the old apparatus completion is obstructed; the two-phase causal parent loses its clock on the fixed-charge quotient	Audit the unrestricted charged-D branch, then build an action-derived observable only if its clock survives the full reduction.
Black-hole radiation	Static theorem and horizon reach in both parities; finite norm at infinity selects Einstein in both parities at one fixture	Generalize in frequency and ℓ, then construct the asymptotic phase space.
Einstein–Weyl relative theory	Matching action-derived tensors through q_3; strict f_1 has a nonzero arity-two defect; global five-charge receiver and Hessian Green-current cone certified	Precompose all stabilizers, add cyclic duals, compare with Lee–Wald, and determine the correct relative nonlinear extension.
Residual branch mixing	Retained cyclic representative and rank-46 carrier exist; physical interaction class not certified	Finish the second-jet image test and obtain an admissible branch resolution.
Quantum anomaly and states	Strict anomaly, compensator restoration, reduced Hadamard carrier, and determinant subproblems certified	Assemble the physical Hessian/determinant and test the Lorentzian full-BV QME/state.
Asymptotic Bach/BMS	Programme stage only	Construct differentiable charges, flux, falloffs, and signs at null infinity.

Method and computational accountability

Palm sets the scientific direction and retains editorial responsibility. AI systems propose and execute much of the detailed mathematics, software, review, literature comparison, and drafting. Neither credentials nor fluent output count as evidence. The certified systems range from dozens to hundreds of linked field and gauge components and can contain more than one hundred thousand exact symbolic interaction terms.

Every material claim is expected to provide: a typed scope; exact or controlled inputs; a machine-readable certificate with hashes and unresolved fields; an independent verifier; mutation tests that must fail on damaged inputs; and a human-readable statement of what was and was not established. Convention repairs and withdrawn promotions remain visible. Publication artifacts are rebuilt from pinned sources. This makes the repository a symbolic laboratory rather than a collection of unchecked AI prose.

Technical manuscripts, certificates, and verification reports

Core papers

- Series overview
- Paper 7: residual cohomology
- Paper 8: covariant causal transport
- Papers 7–8 computational supplement
- Paper 9: relational clocks
- Paper 10: compact Einstein–Maxwell/Weyl–Maxwell phase space
- Paper 11: retained mixed gravity–Maxwell tensor
- Paper 12: one-loop BV anomaly
- Paper 13: finite-harmonic second-order tangent cone
- Paper 14: pure-Weyl black-hole radiation

Causal geometry

- Cyclic Green-transfer bridge
- Curved Nariai causal transfer
- Formal transverse Nariai variation
- Free curvature-observable causal propagator
- Free curvature CCR algebra

Einstein/additional comparison

- Additional-branch current bridge
- Generic axial operator report
- Polar operator certificate
- Polar physical module certificate
- Complete noncyclic linear triangle
- Einstein–Maxwell tensors through arity three
- Relative arity-two defect
- Relative five-charge receiver
- Why the local receiver must be a current
- Polarized relative current seed
- Relative Hessian Green-current cone

Nonlinear tangent cone and interactions

- Charge-fibre bridge
- Complete global plus ell2 bounded cone
- Standard global bounded cone
- Aligned additional bounded obstruction
- Aligned formal secular extension
- All-ell tuned standard cone
- Two-momentum axisymmetric workload
- Two-momentum nonaxisymmetric L3 matrix
- Two-momentum nonaxisymmetric L1/L3 completion
- Berger first-jet redefinition
- Berger second-jet source
- Rank-46 cyclic branch carrier

Clocks and observers

- Retarded relational Maxwell observable
- Localized emitter rank-two response
- Mass-uniform leading response
- First retarded emitter channel
- 108-row first-jet unary obstruction

Black holes

- Static spherical family
- Normalized generator and first law
- Axial operator
- Axial horizon reach
- Axial horizon pairing
- Scoped axial endpoint analysis
- Axial finite-norm boundary class
- Polar operator
- Polar horizon reach
- Polar finite-norm boundary class

Quantum anomaly and states

- Anomaly coefficients and D descent
- Anomaly-induced action
- Reduced cylinder Hadamard/Krein carrier
- Berger microlocal gate
- Generic ghost Schur reduction
- Product $S^2 \times S^2$ ghost Schur carrier

Release and audit infrastructure

- Live D-quotient ledger
- Residual atlas
- Technical certificate graph
- Publication release audit
- Universe-building roadmap
- General-audience introduction

Detailed changelog

- **19 July 2026:** revised Paper 13 terminology to finite exponential-polynomial continuation; added the complete global plus ($\ell = 2, k = 0$) bounded subcone and retained the open full bounded cone. Narrowed the Schwarzschild result to horizon-and-leading-asymptotic nonselection, added polar horizon reach, and updated observer and one-loop determinant work. Corrected the Berger mixed-interaction status to removal through first jet with the second-jet image gate open.
- **18 July 2026:** added the noncyclic Einstein–Weyl linear triangle, Einstein–Maxwell tensors through q_3, the formal Nariai causal variation, black-hole horizon pairing fixtures, reduced cylinder Hadamard/Krein signs, and the first bounded/secular nonlinear classifications.
- **17 July 2026:** added the retarded Berger redshift observable, mixed q_3, retained ℓ_3, rank-46 carrier, localized detector preparations, and the corrected nonlinear generator $K_{\mathrm{Berger}} = D - \omega R$.
- **16 July 2026:** created the physicist-facing summary; recorded the cylinder causal bridge, sector-dependent D, generic axial strict inclusion, and fail-closed Berger microlocal promotion.

Historical entries record what was known at the time. The scientific sections and result ledger above are authoritative when an older phrase differs.

Update policy

This remains the programme's single physicist front door. A new result changes the visible narrative only when a lifecycle advances or retreats, a physical dependency changes, a new background or boundary changes a verdict, a decisive open test closes, or a manuscript changes publication state. Raw term counts and implementation progress remain in team reports.

Each update must preserve one authoritative wording per result, place its strongest limitation beside it, keep distinct backgrounds separate, and retain the technical evidence and history inside this same document.